Panoptic MapAnything

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Abstract

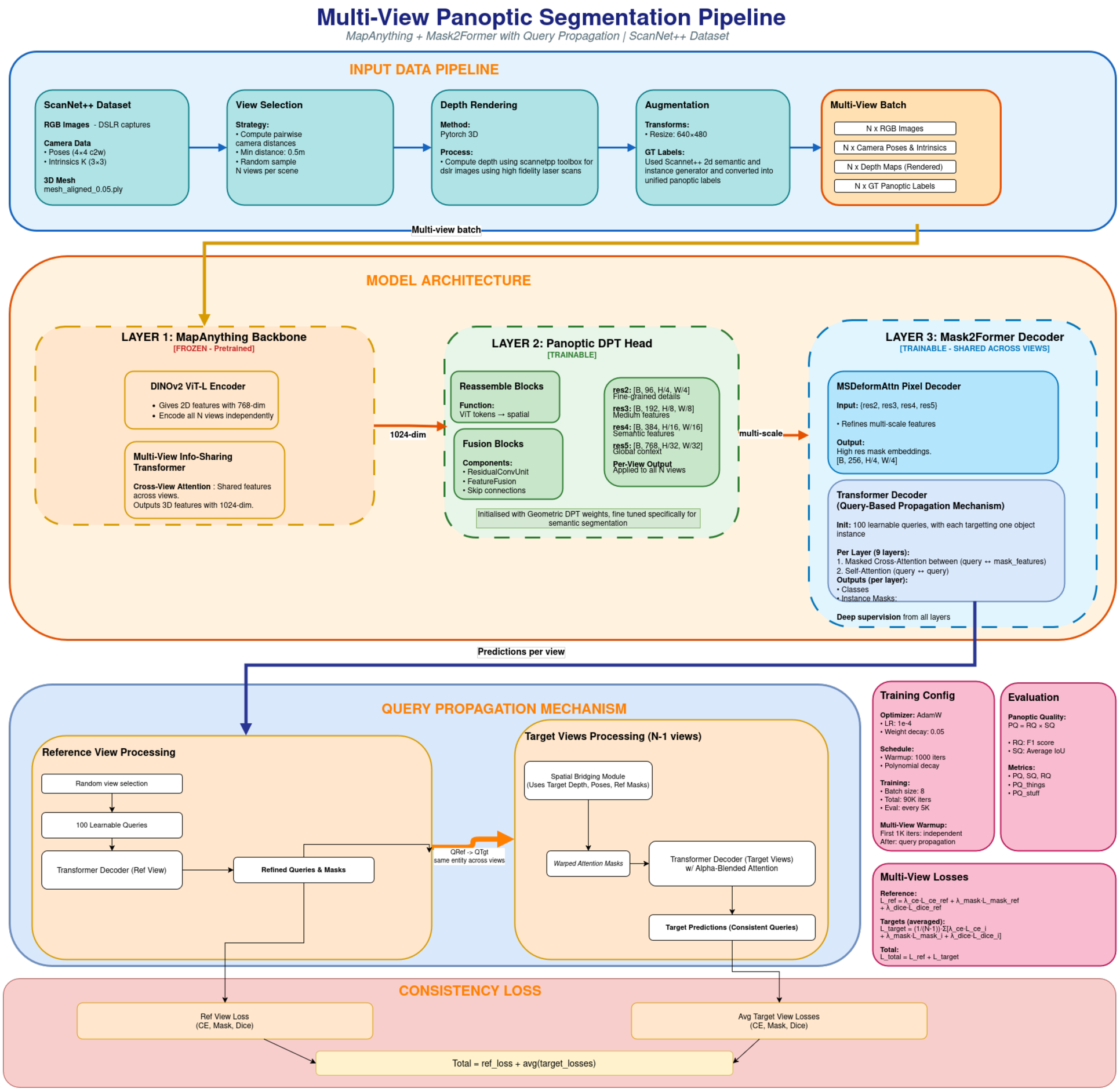
Current 3D panoptic scene segmentation approaches rely on specific inputs such as point clouds, posed rgb images and they demand costly test time optimization for 2D-3D alignment[1]. These models often have seperate segmentation and reconstruction pipelines with additional post processing steps which results in additional overhead with efficiency losses. Our approach mitigates these deficits by effectively utilising a 3D reconstruction backbone model such as MapAnything[2] with a customised Mask2former[3] head for multi view panoptic segmentation thereby creating a unified metric accurate feature rich semantic 3D map. We are using the ScanNet++[4] dataset which is a large scale benchmark dataset. It provides a sub-millimeter resolution laser scans with RGB-D streams for developing richer 3D panoptic segmentation and Novel view synthesis models. We believe that our approach would show that the multiview features from the mapanything backbone can be effectively used by mask2former to provide a multi view consistent panoptic segmentation compared to other approaches with less overhead.

Methodology

Our approach involved the following steps:

1. **Data Acquisition and Multiview Sampling:** Scannet++ Dataset[4] - Preprocessed depth information and then created coco based panoptic labels. Multi-view sampling is done based on minimum camera threshold.
2. **Training Architecture:**
   - Frozen MapAnything Backbone[2] which uses 768-dim DINOV2 ViT-L encoder and multi-view info sharing transformer to generate multi-view aware features (N, B, C, H/14, W/14).
   - Trainable Panoptic DPT Head initialised with geometric prior to output multi-scale features (res2→res5) at strides 4, 8, 16, 32.
   - Custom Multi View Mask2former[3] with Query Propagation and warped mask attention[5] across reference view and target view which ensures multi view consistency on its panoptic predictions.
3. **Loss Calculation:** Total Loss is a sum of Reference view loss and average of target view losses.
4. **Evaluation:** PQ,SQ,RQ metrics using COCO Panoptic Evaluator

Pipeline Overview



Challenges faced:

- Lack of official guidelines on the exact number of semantic and instance classes in scannet++ dataset, made the preprocessing and dataloading steps such as panoptic label generation & rendering depth information challenging.
- Due to unavailabity of A100 GPU cluster, we enabled gradient checkpointing which increased training time.

Future Work:

- To unfreeze the mapanything backbone for training and then evaluate whether it depicts better loss convergence and fixes the vanishing gradient issue in mask loss.
- To integrate Mask2former head with open vocabulary approaches like SigLip for better panoptic labelling of scenes.

References

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